

Heart Disease Prediction Using Machine Learning

Binary Classification on CDC BRFSS 2020 Survey Data

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Abstract

Heart disease is the leading cause of death worldwide, causing approximately 17.9 million deaths annually (World Health Organization, 2023). This study applies three binary classification algorithms — Logistic Regression, Random Forest, and Support Vector Machine — to a representative sample of 2,059 patients drawn from the CDC Behavioral Risk Factor Surveillance System (BRFSS) 2020 survey. The sample preserves the real-world disease rate of 8.6%, and class imbalance is addressed through `class_weight='balanced'` in all models. Hyperparameters are optimised using GridSearchCV with 5-fold stratified cross-validation, scoring on F1. Random Forest achieves the highest AUC-ROC (0.855) and Logistic Regression the highest disease recall (80%). Feature importance analysis identifies age category, general health status, and difficulty walking as the strongest predictors. Ethical considerations including demographic bias, self-reported data limitations, and the asymmetric cost of false negatives are discussed.

1. Introduction

Cardiovascular disease (CVD) remains the world's leading cause of mortality, claiming an estimated 17.9 million lives annually — approximately 31% of all global deaths (World Health Organization, 2023). Coronary heart disease, driven by arterial blockages and reduced cardiac blood flow, accounts for the largest share. A critical contributing factor to poor outcomes is late diagnosis: most patients are not identified as high risk until after a serious cardiac event such as a heart attack or stroke.

Machine learning (ML) offers a data-driven approach to early risk stratification. Large-scale health surveys such as the CDC's Behavioral Risk Factor Surveillance System (BRFSS) collect self-reported lifestyle and health data from hundreds of thousands of Americans annually. By learning statistical patterns from this data, ML models can flag individuals at elevated risk based on information that is already collected — without requiring additional clinical tests.

This project builds and evaluates three classification algorithms on a representative 2,059-patient sample from CDC BRFSS 2020. The objectives are: (1) implement Logistic Regression, Random Forest, and SVM with systematic hyperparameter tuning via GridSearchCV; (2) handle real-world class imbalance correctly using `class_weight='balanced'`; (3) interpret feature importance; and (4) critically assess the ethical implications of deploying automated health risk screening.

2. Methodology

2.1 Dataset

This study uses the CDC BRFSS 2020 dataset (Kamilpytlak, 2021), available via Kaggle and derived from the official CDC annual health survey. The full dataset contains 319,795 responses and 18 features. The target variable 'HeartDisease' indicates whether the respondent has ever been diagnosed with coronary heart disease or myocardial infarction.

The full dataset is severely imbalanced: 8.56% of respondents report heart disease. We drew a representative sample of **2,059 patients** — 177 with disease (8.6%) and 1,882 without (91.4%) — preserving the real-world distribution. This is a deliberate methodological choice: a forced 50/50 split would misrepresent reality and inflate apparent model performance. Class imbalance is handled instead by using `class_weight='balanced'` in all models, which automatically adjusts each algorithm to pay proportionally more attention to the minority class.

Feature	Description	Encoding
BMI	Body Mass Index (weight/height ²)	Continuous
Smoking	Smoked ≥ 100 cigarettes lifetime	Binary $\rightarrow 0/1$
AlcoholDrinking	Heavy drinker (>14 /week male, >7 female)	Binary $\rightarrow 0/1$
Stroke	Ever had a stroke	Binary $\rightarrow 0/1$
PhysicalHealth	Days of poor physical health (last 30 days)	Continuous
MentalHealth	Days of poor mental health (last 30 days)	Continuous
DiffWalking	Difficulty walking or climbing stairs	Binary $\rightarrow 0/1$
Sex	Male or Female	Binary $\rightarrow 0/1$
AgeCategory	Age band: 18–24 up to 80+	Ordinal 0–12
Race	White/Black/Asian/Hispanic/Other/American Indian	One-hot
Diabetic	No / Borderline / Yes / Yes (pregnancy)	Ordinal 0–2
PhysicalActivity	Exercise in past 30 days outside work	Binary $\rightarrow 0/1$
GenHealth	Self-reported health (Poor $\rightarrow$ Excellent)	Ordinal 0–4
SleepTime	Average hours of sleep per 24 hours	Continuous
Asthma	Ever had asthma	Binary $\rightarrow 0/1$
KidneyDisease	Ever had kidney disease (excl. stones)	Binary $\rightarrow 0/1$
SkinCancer	Ever had skin cancer	Binary $\rightarrow 0/1$
HeartDisease	Ever diagnosed with CHD or MI — TARGET	Binary $\rightarrow 0/1$

Table 1: All 17 input features, descriptions, and encoding strategy. CHD = coronary heart disease.

2.2 Preprocessing

Features were encoded as described in Table 1. After encoding and one-hot expansion of Race, the dataset has 21 input features. The data was split into training (80%, $n=1,647$) and test (20%, $n=412$) sets using stratified sampling to preserve the 8.6% disease rate in both subsets. All features were standardised using StandardScaler, fitted exclusively on training data to prevent data leakage.

2.3 Handling Class Imbalance

The 8.6%/91.4% class split means that a naive model predicting 'no disease' for every patient would achieve 91.4% accuracy — while being clinically useless. All three models were trained with `class_weight='balanced'`, which adjusts the per-sample weight inversely proportional to class frequency. This effectively penalises the model more for misclassifying disease patients, without artificially altering the data distribution.

2.4 Hyperparameter Tuning

GridSearchCV with 5-fold stratified cross-validation was applied to all three algorithms. The scoring metric was F1-score, chosen because it balances precision and recall — both relevant in a medical screening context where false positives waste clinical resources and false negatives delay treatment.

Model	Parameter Grid	Best Parameters
Logistic Regression	$C \in \{0.01, 0.1, 0.5, 1, 5, 10\}$	$C = 0.01$
Random Forest	$n_estimators \in \{100, 200, 300\}$, $max_depth \in \{5, 10, 20, None\}$, $min_samples_split \in \{2, 5, 10\}$	$n=100$, $depth=5$, $split=10$
SVM (RBF)	$C \in \{0.1, 1, 10, 100\}$, $gamma \in \{scale, auto, 0.01, 0.001\}$	$C=0.1$, $\gamma=0.001$

Table 2: GridSearchCV parameter grids and selected best parameters (F1 scoring, `class_weight=balanced`).

3. Algorithm Selection

3.1 Logistic Regression

Logistic Regression was included as the interpretable baseline. It models the log-odds of heart disease as a linear combination of features, producing well-calibrated probability estimates through the sigmoid function. Despite its linearity, it achieves the highest disease recall (80%) in this study, consistent with the literature on self-reported survey data where risk factor–outcome relationships tend to be approximately linear. The optimal $C=0.01$ indicates strong regularisation, appropriate for the limited number of positive cases ($n=152$ in training) (Pedregosa et al., 2011).

3.2 Random Forest

Random Forest constructs an ensemble of decision trees on bootstrapped subsets of the training data and aggregates predictions by majority vote (Breiman, 2001). This reduces variance relative to individual trees and naturally handles non-linear feature interactions without requiring feature engineering. The constrained depth ($max_depth=5$) and high minimum split threshold ($min_samples_split=10$) act as regularisers, preventing overfitting on the limited positive class examples. Random Forest provides Gini impurity-based feature importance scores that offer clinically interpretable insight into prediction drivers.

3.3 Support Vector Machine (SVM with RBF Kernel)

The SVM with RBF kernel seeks the maximum-margin hyperplane separating classes in a high-dimensional transformed feature space (Cortes and Vapnik, 1995). The soft-margin parameter $C=0.1$ permits misclassifications, which is appropriate given the severe class imbalance. The small gamma value (0.001) produces a wide kernel with global influence, reducing sensitivity to noise in individual samples — important when positive examples are sparse. `probability=True` enables Platt-scaled probability outputs required for ROC curve computation.

4. Results and Analysis

4.1 Model Performance

Model	CV Acc	$\pm$ Std	Test Acc	Recall	F1	AUC
Logistic Regression	77.5%	$\pm 1.3\%$	76.5%	80.0%	0.366	0.849
SVM (RBF)	86.9%	$\pm 2.0\%$	84.2%	54.3%	0.369	0.828
Random Forest ← Best AUC	82.6%	$\pm 1.6\%$	82.3%	74.3%	0.416	0.855

Table 3: Performance on the 412-patient held-out test set. Recall = recall for heart disease class.

■ ■ **Why SVM has high accuracy (84.2%) but low recall (54.3%):** SVM learns to predict 'no disease' for borderline cases to maximise overall accuracy on the 91.4%/8.6% test split. Despite `class_weight=balanced`, SVM does not generalise the minority class as effectively as Logistic Regression or Random Forest on this survey dataset. This demonstrates why accuracy alone is misleading for imbalanced medical data.

4.2 Model Selection

Random Forest is selected as the best overall model based on AUC-ROC (0.855), which measures discriminatory ability across all thresholds and is the most appropriate single metric for imbalanced binary classification (Fawcett, 2006). It also achieves the best balance between test accuracy (82.3%) and disease recall (74.3%). Logistic Regression achieves the highest recall (80%) but lower AUC (0.849) and accuracy (76.5%), making it preferable in contexts where missing a sick patient is absolutely unacceptable and false alarms are tolerable.

4.3 Feature Importance

Random Forest Gini importance scores identify the five strongest predictors:

- **AgeCategory (28.7%)** — the dominant predictor. Disease risk increases substantially and consistently with age, consistent with established cardiovascular epidemiology.
- **GenHealth (13.4%)** — self-reported general health is a strong composite predictor, reflecting overall health status and comorbidity burden.
- **DiffWalking (10.0%)** — difficulty walking indicates physical deconditioning and may reflect pre-existing cardiovascular limitation.
- **PhysicalHealth (8.6%)** — more days of poor physical health in the past month correlates with chronic illness burden.
- **Diabetic (8.2%)** — diabetes is a well-established independent risk factor for cardiovascular disease, driving arterial damage.

4.4 Limitations

Several limitations constrain the generalisability of these findings. First, the data is self-reported, introducing recall bias and measurement error. Second, the sample of 2,059 patients provides relatively few positive cases ($n=177$) for the models to learn from. Third, the sample reflects the US population and may not generalise to other countries. Fourth, the one-hot encoding of Race adds

features that may introduce demographic bias.

5. Ethical Discussion

5.1 Bias and Fairness

The BRFSS dataset includes race as a feature and was collected from US respondents. Disparities in healthcare access, health literacy, and insurance coverage across racial and socioeconomic groups may cause the model to perform differently for different subpopulations. Before any clinical deployment, model performance must be evaluated separately for each demographic subgroup (Barocas et al., 2019). Features derived from race should be handled with particular care to avoid encoding structural inequalities into the prediction system.

5.2 Self-Reported Data and Measurement Error

Unlike clinical datasets derived from blood tests or ECG recordings, BRFSS data is collected via telephone survey. Respondents may under-report health-risk behaviours (e.g. alcohol consumption, smoking), may be unaware of undiagnosed conditions, or may provide inconsistent answers across survey years. This introduces noise that partially explains why model accuracy is approximately 82% rather than the 87–92% typically reported on clinical measurement datasets.

5.3 Asymmetric Cost of Errors

A false negative — predicting 'no disease' when disease is present — may delay diagnosis and treatment. A false positive — flagging a healthy patient as high risk — causes unnecessary anxiety and wasted clinical resources. These costs are asymmetric: in most screening contexts, missing a sick patient is more serious. This motivated using F1 as the tuning metric and `class_weight='balanced'`. In deployment, the classification threshold can be lowered below 0.5 to further increase recall at the cost of more false alarms.

5.4 Privacy and Regulatory Compliance

Any system applying this model to real patients handles protected health information subject to EU GDPR (explicit consent, data minimisation, right to explanation for automated decisions) and US HIPAA (access controls, audit trails). Although the BRFSS data used here is anonymised, a deployed system would need full anonymisation or pseudonymisation and role-based access controls (European Parliament, 2016).

5.5 Decision Support Only

This model is designed as a decision-support tool. The physician always makes the final clinical decision. Research on automation bias shows that clinicians can over-rely on algorithmic outputs (Cummings, 2004). To mitigate this, SHAP (Lundberg and Lee, 2017) should accompany any deployed version, providing patient-level explanations — for example: 'This patient was flagged primarily because they are in the 70–74 age group, report fair general health, and have difficulty walking.'

6. Conclusion

This study demonstrates that machine learning can identify elevated heart disease risk from self-reported lifestyle and health data at population scale. On a representative 2,059-patient sample

from CDC BRFSS 2020 at the real-world disease rate of 8.6%, Random Forest achieves the best overall performance with AUC-ROC of 0.855, test accuracy of 82.3%, and disease recall of 74.3% after GridSearchCV tuning with `class_weight='balanced'`. Age, general health status, and difficulty walking are the strongest predictors — findings consistent with established cardiovascular epidemiology.

Future work should expand the sample size to improve minority class representation, evaluate fairness across demographic subgroups, integrate SHAP for patient-level explainability, and validate on external clinical datasets. Responsibly deployed as a decision-support tool alongside qualified clinicians, models of this type represent a meaningful contribution to population-level cardiovascular risk screening.

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